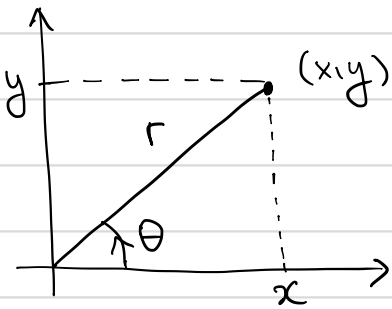
Polar coordinates and applications of $\iint$ (§15.3-4)

Polar coordinates

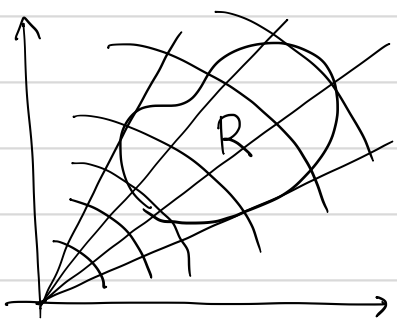


$x = r\cos\theta$

$y = r\sin\theta$

$r = \sqrt{x^2+y^2}$

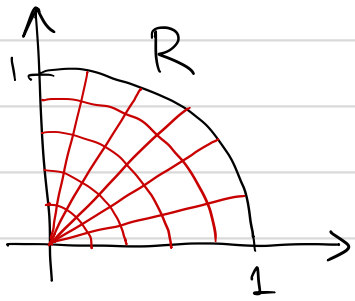
$\tan\theta = y/x$



$$\iint_R f(x,y)\,dA =$$

$$= \int_{\theta_{min}}^{\theta_{max}} \int_{r_{min}(\theta)}^{r_{max}(\theta)} f(r\cos\theta, r\sin\theta)\, \underline{\underline{r}}\,dr\,d\theta.$$

Example: $f(x,y) = 1 - x^2 - y^2$



$x^2+y^2 \leq 1$

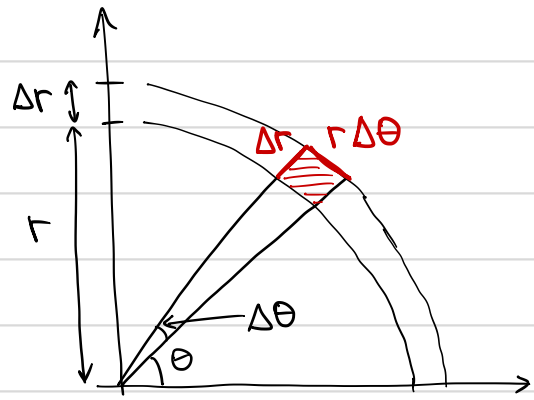
$x \geq 0,\ y \geq 0$

$$\iint_R 1-x^2-y^2\,dA =$$

$$= \int_0^{\frac{\pi}{2}}\int_0^1 (1-r^2)\underbrace{r\,dr\,d\theta}_{dA}$$

$$= \int_0^{\pi/2}\left[\frac{r^2}{2} - \frac{r^4}{4}\right]_{r=0}^{1} d\theta = \frac{1}{4}\cdot\frac{\pi}{2} = \frac{\pi}{8}$$

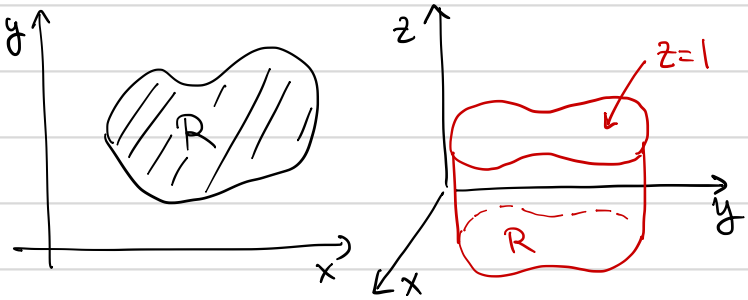
Why $dA = r\,dr\,d\theta$:



Applications:

1) Find area of region R:

$$\text{Area}(R) = \iint_R 1 \cdot dA$$



2) Mass of planar object with density ρ = mass per unit area

$$\Delta m = \rho \, \Delta A$$

$$\text{Mass} = \iint_R \rho(x,y) \, dA$$

3) Average value of f in R

$$(\text{Average of } f) = \frac{1}{\text{Area}} \iint_R f \, dA$$

$$\begin{pmatrix}\text{Weighted} \\ \text{average of } f\end{pmatrix} = \frac{1}{\text{Mass}} \iint_R f \rho \, dA$$

Sanity check:

If $m = \min\{f(x,y) \mid (x,y) \in R\}$

$M = \max\{f(x,y) \mid (x,y) \in R\}$

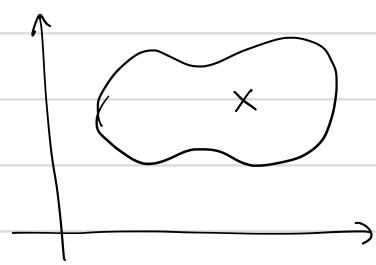
then

$$m \leq \overbrace{\frac{1}{\text{Area}} \iint_R f \, dA}^{\text{average}} \leq M$$

since

$$\underbrace{\iint_R m \, dA}_{m \cdot \text{Area}(R)} \leq \iint_R f \, dA \leq \underbrace{\iint_R M \, dA}_{M \cdot \text{Area}(R)}$$

4) Center of mass of a planar object (with density ρ): $(\bar{x}, \bar{y})$



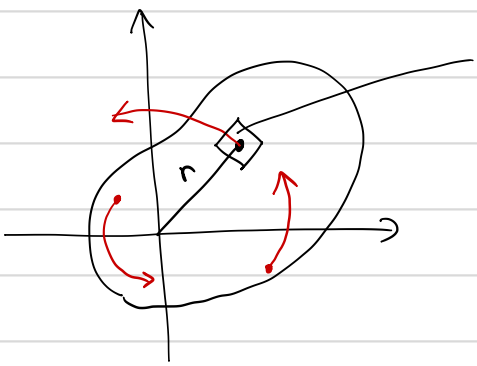
$$\bar{x} = \frac{1}{\text{Mass}} \iint_R x \, \rho(x,y) \, dA$$

$$\bar{y} = \frac{1}{\text{Mass}} \iint_R y \, \rho(x,y) \, dA$$

5) Moment of inertia

mass = resistance to translation

moment of inertia about an axis = resistance to rotation about that axis.



$\Delta m = \rho \Delta A$

ω = angular speed (rad/s)

$v = \omega r$

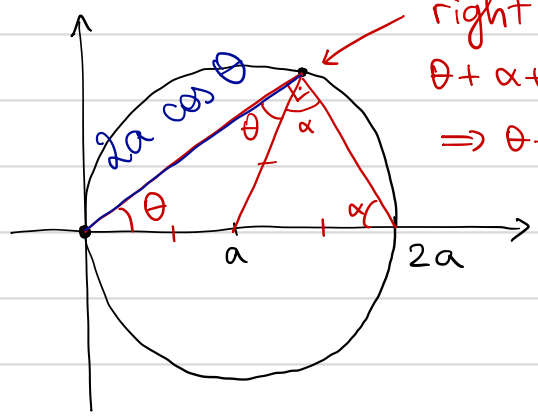
$$K.E. = \frac{1}{2}(\Delta m) v^2$$
$$= \frac{1}{2}(\Delta m)(\omega r)^2$$
$$= \frac{1}{2}\underbrace{(r^2 \rho \Delta A)}_{\text{inertia}} \omega^2$$

Moment of inertia about the origin:

$$\boxed{I_0 = \iint_R r^2 \rho \, dA}$$

Rotational kinetic energy is $\frac{1}{2} I_0 \omega^2$.

Example: Moment of inertia I_0 of disk about point in the circle. (Density: $\rho = 1$.)



right angle:

$\theta + \alpha + (\theta + \alpha) = \pi$

$\Rightarrow \theta + \alpha = \pi/2$.

skip in class.

$$I_0 = \iint r^2 \rho \, dA$$
$$= \int_{-\pi/2}^{\pi/2} \int_0^{2a\cos\theta} r^2 \cdot r \, dr \, d\theta$$
$$= \int_{-\pi/2}^{\pi/2} \left[\frac{r^4}{4} \Big|_0^{2a\cos\theta} \right] d\theta$$
$$= 4a^4 \int_{-\pi/2}^{\pi/2} \cos^4\theta \, d\theta$$
$$= \cdots = \frac{3}{2}\pi a^4$$

Remember: $\cos^2\theta = \frac{1}{2}(\cos(2\theta) + 1)$

Alternative to trigonometry:
Find equation in xy-coords and translate to polar coords:

$$(x-a)^2 + y^2 = a^2$$
$$\Leftrightarrow x^2 - 2ax + a^2 + y^2 = a^2$$
$$\Leftrightarrow x^2 + y^2 = 2ax$$
$$\Leftrightarrow r^2 = 2ar\cos\theta$$
$$\Leftrightarrow r = 2a\cos\theta \qquad (r \neq 0)$$